

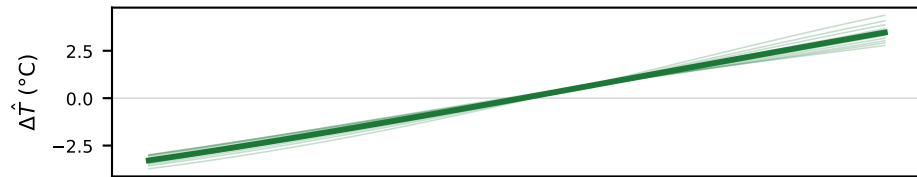
The Fidelity-Utility Paradox in Surrogate-Based RL for HVAC Control

A more accurate surrogate exposes a rougher action $\rightarrow$ temperature response (A); a policy trained on it collapses on the live building (B).

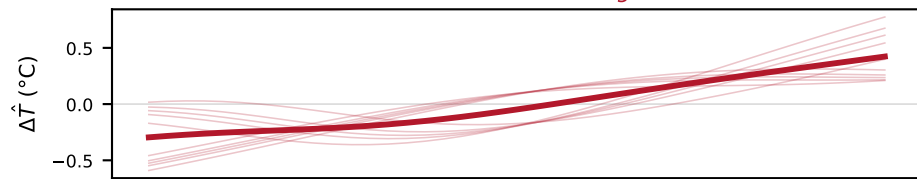
A role-separating hybrid keeps the response smooth and the controller deployable.

(A) Measured response surface

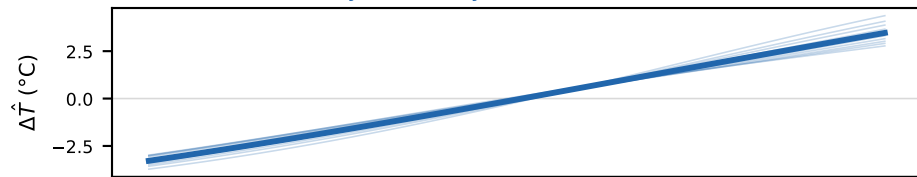
v3 (coarse) — smooth



accurate twin — rough



hybrid (v3 dynamics) — smooth



(B) Fidelity-utility deployment plane (measured)

